



# UNIVERSIDAD DE GRANADA

Faculty of Science

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## Monte Carlo Simulation of Dose Deposition in Mammography

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# Contents

|                                                         |          |
|---------------------------------------------------------|----------|
| <b>Mammographic Dose Simulation</b>                     | <b>2</b> |
| 1 Theoretical Background . . . . .                      | 2        |
| 2 Methods and Setup . . . . .                           | 2        |
| 2.1 Geometry . . . . .                                  | 2        |
| 2.2 Materials . . . . .                                 | 3        |
| 2.3 X-ray spectra . . . . .                             | 4        |
| 2.4 Simulation settings . . . . .                       | 5        |
| 3 Energy Deposition for the Three Spectra . . . . .     | 5        |
| 4 Depth-Dose Distribution and Integrated Dose . . . . . | 7        |
| 4.1 Depth-dose profile and 2D dose map . . . . .        | 7        |
| 4.2 Total dose to the breast . . . . .                  | 8        |
| 5 Effect of the Glandular Fraction . . . . .            | 9        |
| 6 Conclusions . . . . .                                 | 11       |

# Mammographic Dose Simulation

## 1. Theoretical Background

Mammography is a low-energy X-ray imaging modality used to detect small contrast differences between glandular and adipose tissue inside the breast. The whole physics of the technique sits in a very narrow energy window, between approximately 10 and 30 keV.

In this energy range the dominant photon-matter interaction in soft tissue is the *photoelectric effect*, whose cross-section scales approximately as  $\sigma_{\text{ph}} \propto Z^n/E^3$  with  $n \simeq 4-5$ . The strong dependence on the atomic number  $Z$  is what produces the contrast between glandular tissue ( $Z_{\text{eff}} \simeq 7.4$ ) and adipose tissue ( $Z_{\text{eff}} \simeq 6.3$ ); above  $\sim 30$  keV Compton scattering takes over and this contrast collapses, which is why mammographic tubes are operated at 25–30 kVp rather than at higher energies.

The quantity of interest is the absorbed dose  $D$ , defined as the energy deposited per unit mass of irradiated material:

$$D = \frac{dE}{dm} \quad [\text{J/kg} = \text{Gy}]. \quad (1)$$

The Monte Carlo code transports each primary photon through the geometry, samples its interactions and tallies the energy deposited in each volume element. The output is therefore the energy per primary photon (eV/g per simulated history); to obtain a physical dose value the result must be scaled by the actual number of primary photons emitted by the tube during the exposure, i.e. by the integral of the beam current over the exposure time.

Two clinically relevant quantities are evaluated in this practice: the *skin dose*, related to the photons absorbed in the thin 0.2 cm epithelial layer that surrounds the breast, and the *average breast dose*, integrated over the cylindrical glandular volume. Selecting an appropriate spectrum amounts to finding a compromise between the contrast it produces (favoured by lower energies) and the dose it deposits in the patient (lower for harder beams).

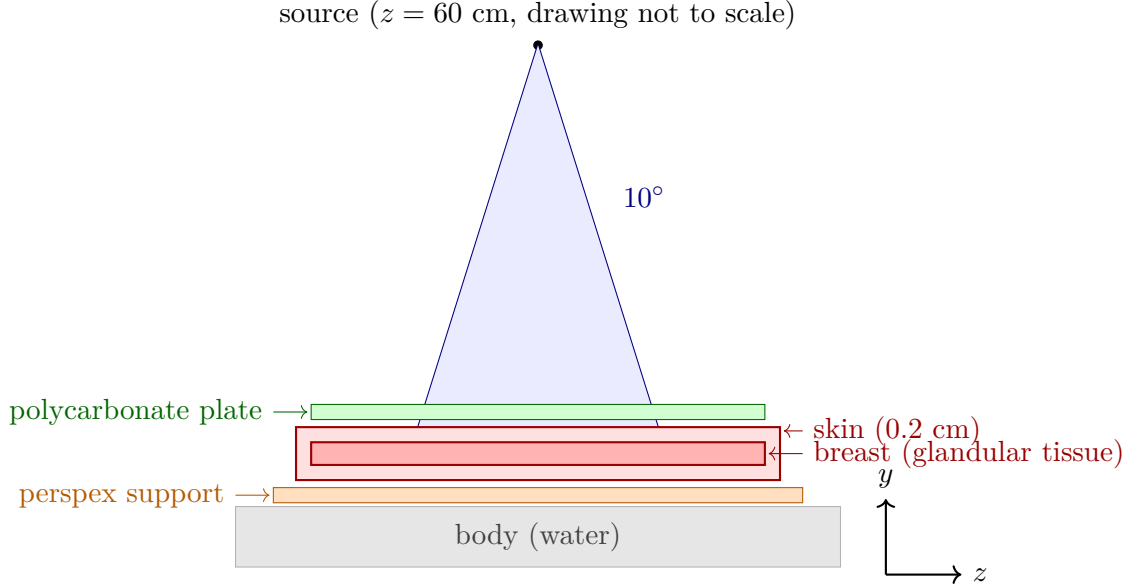
## 2. Methods and Setup

The Monte Carlo simulations are run with **penmain** from the PENELOPE-2018 distribution. The geometry, the absorber materials, the input file template and the spectra are those provided with the practice.

### 2.1. Geometry

Figure 1 shows a schematic of the simulation geometry in the  $yz$  plane (lateral view). The X-ray source is a point isotropic photon emitter located at  $z = 60$  cm above the breast, irradiating downwards through a conical beam with  $10^\circ$  aperture. The breast is modelled as a half-cylinder of radius  $r = 8$  cm, lying with its cylindrical axis along the  $y$ -direction, so that the beam crosses it through its diameter; the patient's body fills the half-space behind it ( $y < 0$  in the lab frame after the geometry rotation). A 0.2 cm thick *skin* layer of epithelial tissue surrounds the breast

on all sides. The cylinder length is  $h = 6$  cm in the original PENELOPE template (`mama.geo`), used in Section 3 (Punto 1), and  $h = 5$  cm in the modified geometry (`mama_1.geo`) used in Sections 4 and 5 (Punto 2 and 3) as required by the assignment. A polycarbonate compression plate and a perspex support plate sit above and below the breast respectively, while the rest of the volume is filled with air. A semi-infinite slab represents the patient's body.



**Figure 1:** Schematic of the simulation geometry in the  $yz$  lateral plane (drawing not to scale). The X-ray source sits at  $z = 60$  cm on the beam axis; a  $10^\circ$  conical beam illuminates downward the breast cylinder, surrounded by a thin skin layer, sandwiched between a polycarbonate compression plate and a perspex support, with the patient's body occupying the half-space below.

## 2.2. Materials

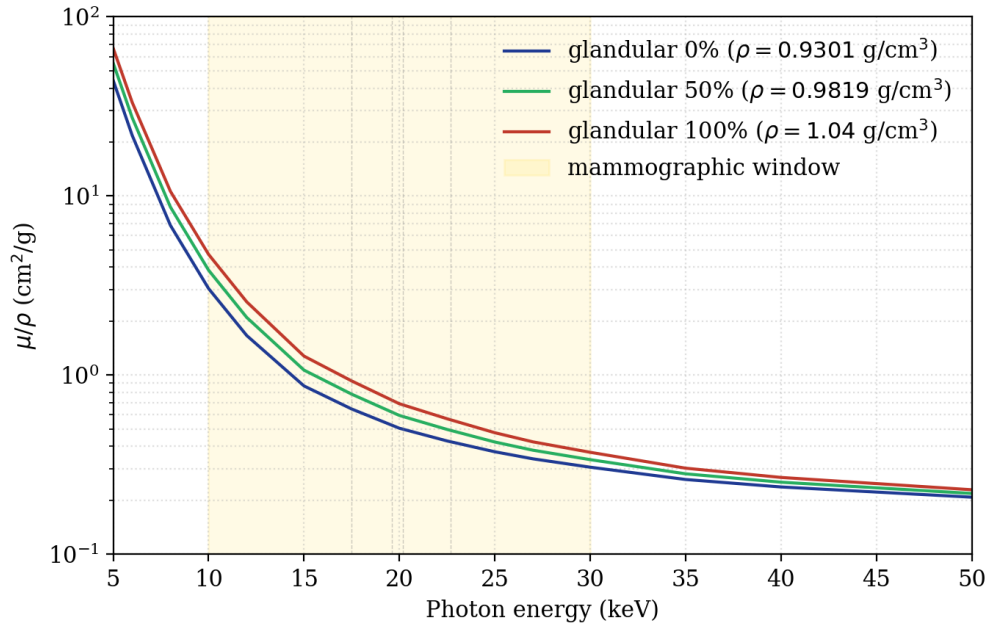
Six materials enter the simulation: *glandular tissue* at three glandularity fractions (0%, 50%, 100%), *epithelial* tissue (skin), *water*, *perspex* (compression plate), *polycarbonate* (support plate) and *air*. Their compositions and densities are taken from the table provided with the practice and reported in Table 1.

| Material                 | $\rho$ (g/cm <sup>3</sup> ) | Composition (mass fractions)                          |
|--------------------------|-----------------------------|-------------------------------------------------------|
| Air                      | 0.00120                     | N (1.00), O (0.27), Ar (0.006), C (0.0002)            |
| Water                    | 0.99821                     | H (1.00), O (0.50)                                    |
| Epithelial tissue (skin) | 1.10000                     | H, C, N, O, Na, Mg, P, S, Cl, K, Fe, Ca, Zn (full)    |
| Glandular 0%             | 0.93010                     | H (1.00), C (0.464), N (0.011), O (0.141), P (0.0003) |
| Glandular 50%            | 0.98190                     | H (1.00), C (0.314), N (0.017), O (0.273), P (0.0009) |
| Glandular 100%           | 1.04000                     | H (1.00), C (0.151), N (0.002), O (0.418), P (0.002)  |
| Perspex                  | 1.19000                     | H (1.00), C (0.625), O (0.250)                        |
| Polycarbonate            | 1.20000                     | H (0.875), C (1.000), O (0.188)                       |

Table 1: Materials used in the simulations, with their density and mass-fraction composition.

The mass attenuation coefficient  $\mu/\rho$  of the breast tissue, computed from the PENELOPE atomic data, is shown in Figure 2. In the mammographic energy window (10–30 keV) it spans almost two decades, from  $\sim 5$  cm<sup>2</sup>/g at the lower end to  $\sim 0.4$  cm<sup>2</sup>/g near the upper end. This

rapid variation is the reason why such a thin energy band has such a large impact on the dose distribution.



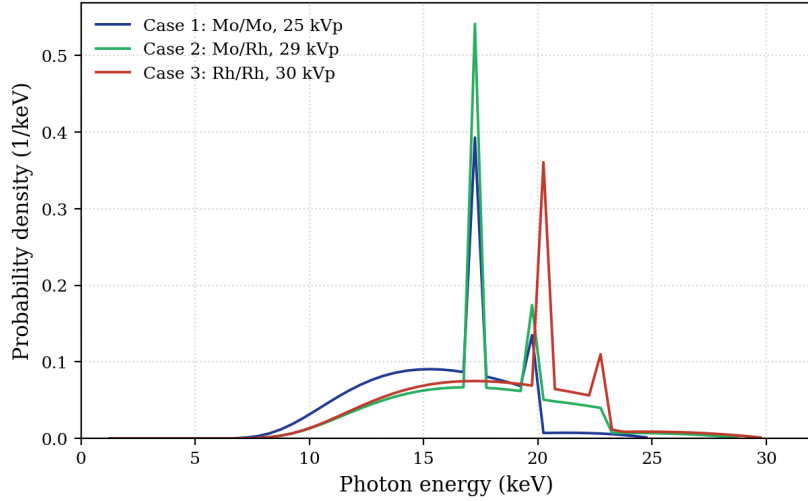
**Figure 2:** Photon mass attenuation coefficient  $\mu/\rho$  for the three breast compositions used in this practice, computed by the mixing rule from the elemental NIST cross-sections. The shaded band indicates the mammographic energy window (10–30 keV); the curves of the three tissues are very close in  $\mu/\rho$ , but the linear attenuation coefficient  $\mu = (\mu/\rho) \cdot \rho$  scales more strongly because of the density difference between the three tissues (0.93 to 1.04 g/cm³).

### 2.3. X-ray spectra

Three spectra typical of mammography practice are considered, all generated with the analytical toolkit SPEKPY v2:

- **Case 1** – Mo target with 30  $\mu\text{m}$  Mo filtration, 25 kVp; representative of thin breasts ( $\sim 4$  cm).
- **Case 2** – Mo target with 25  $\mu\text{m}$  Rh filtration, 29 kVp; intermediate, used for breasts of 5–7 cm.
- **Case 3** – Rh target with 25  $\mu\text{m}$  Rh filtration, 30 kVp; harder beam used for thicker ( $> 7$  cm) breasts.

The three spectra are shown together in Figure 3. The discrete K-line peaks in each spectrum reflect the characteristic radiation of the anode (Mo lines at 17.5 and 19.6 keV; Rh lines at 20.2 and 22.7 keV); the K-edges of the filters introduce a sharp absorption notch that shapes the high-energy tail.



**Figure 3:** The three SpekPy spectra used as primary sources in the simulations: Case 1 (Mo/Mo, 25 kVp), Case 2 (Mo/Rh, 29 kVp) and Case 3 (Rh/Rh, 30 kVp). All spectra are normalised to unit area for shape comparison; the discrete K-line peaks of Mo (17.5 and 19.6 keV) and Rh (20.2 and 22.7 keV) dominate the soft bremsstrahlung continuum.

## 2.4. Simulation settings

For each configuration `penmain` is run with absorption energies set to 1 keV (electrons, photons and positrons), and a 3D dose grid covering the breast volume with  $100 \times 100 \times 300$  voxels. The simulations are stopped once approximately  $10^8$  primary histories have been transported, which is sufficient to obtain  $\lesssim 5\%$  statistical uncertainty on the integrated breast dose. The output files of interest are `depth-dose.dat` (1D profile along the beam axis), `3d-dose-map.dat` (full 3D dose map, used for the integrated dose calculation) and `energy-up.dat/energy-down.dat` (spectra of upward and downward emerging particles). Five PENELOPE runs are performed in total, summarised in Table 2.

| Run | Spectrum      | Glandularity | Used for                     |
|-----|---------------|--------------|------------------------------|
| 1   | Mo/Mo, 25 kVp | 50%          | Section 3 (Case 1)           |
| 2   | Mo/Rh, 29 kVp | 50%          | Section 3 (Case 2)           |
| 3   | Rh/Rh, 30 kVp | 50%          | Sections 3 (Case 3), 4 and 5 |
| 4   | Rh/Rh, 30 kVp | 0%           | Section 5                    |
| 5   | Rh/Rh, 30 kVp | 100%         | Section 5                    |

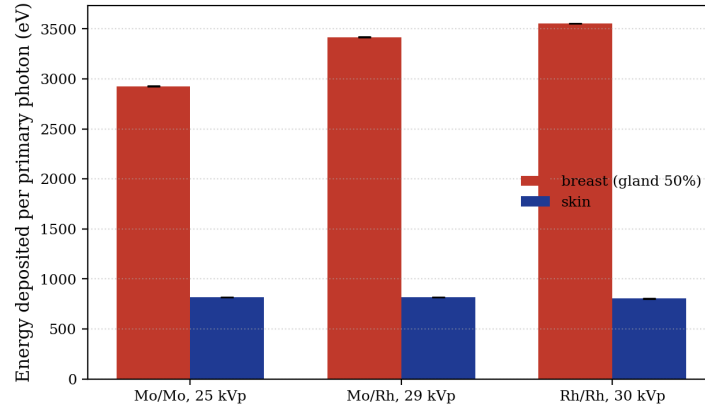
Table 2: The PENELOPE simulation runs performed for the three points of the practice.

## 3. Energy Deposition for the Three Spectra

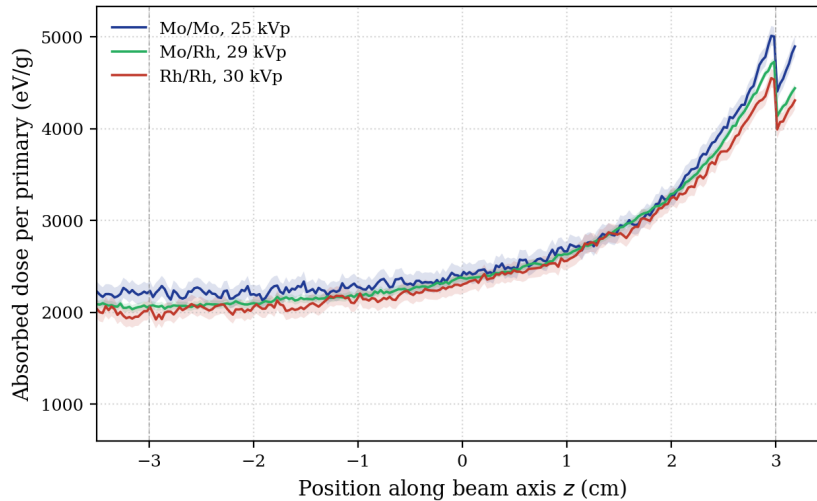
The first part of the practice asks how the energy deposited in the breast and in the skin layer changes when the same phantom (glandular 50%, 5 cm thickness) is irradiated with the three mammographic spectra introduced above. Three observables are compared in this section: the integrated energy absorbed in the glandular volume and in the skin (Table 3 and Figure 4), the depth-dose profiles through the breast (Figure 5) and the 2D dose maps in the  $yz$  plane through the centre of the breast (Figure 6).

| Spectrum               | $\langle E_{\text{breast}} \rangle$ (eV/photon) | $\langle E_{\text{skin}} \rangle$ (eV/photon) | $E_{\text{breast}}/E_{\text{skin}}$ |
|------------------------|-------------------------------------------------|-----------------------------------------------|-------------------------------------|
| Case 1 – Mo/Mo, 25 kVp | $2926.6 \pm 1.3$                                | $817.2 \pm 0.7$                               | 3.58                                |
| Case 2 – Mo/Rh, 29 kVp | $3418.1 \pm 0.7$                                | $816.9 \pm 0.3$                               | 4.18                                |
| Case 3 – Rh/Rh, 30 kVp | $3555.0 \pm 1.6$                                | $803.0 \pm 0.8$                               | 4.43                                |

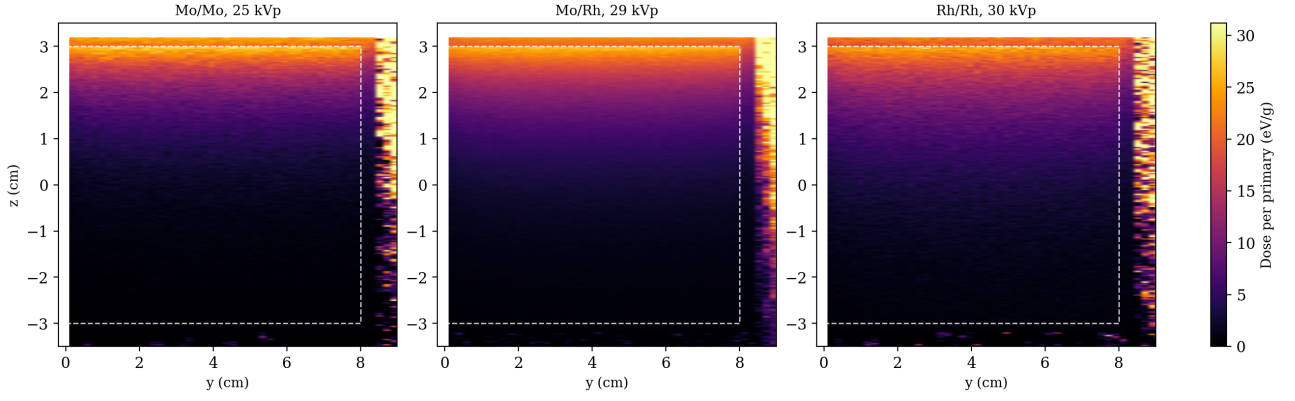
Table 3: Energy deposited per primary photon in the breast (glandular 50%) and in the surrounding skin layer for the three spectra, with the breast-to-skin ratio. Uncertainties are quoted at  $1\sigma$ .



**Figure 4:** Energy deposited per primary photon in the breast (glandular 50%) and in the skin for the three mammographic spectra studied. As the beam hardens (from Case 1 to Case 3) the breast contribution grows from  $\sim 2.9$  to  $\sim 3.6$  keV/photon while the skin contribution stays nearly flat at  $\sim 0.8$  keV/photon.



**Figure 5:** Depth-dose profiles along the beam axis ( $z$ ) inside the breast (glandular 50%) for the three spectra. The beam enters the breast at  $z = +3$  cm and the dose decreases monotonically with depth; the harder Rh/Rh beam shows the slowest decrease, the softer Mo/Mo beam the steepest. Shaded bands are the  $1\sigma$  statistical uncertainty.



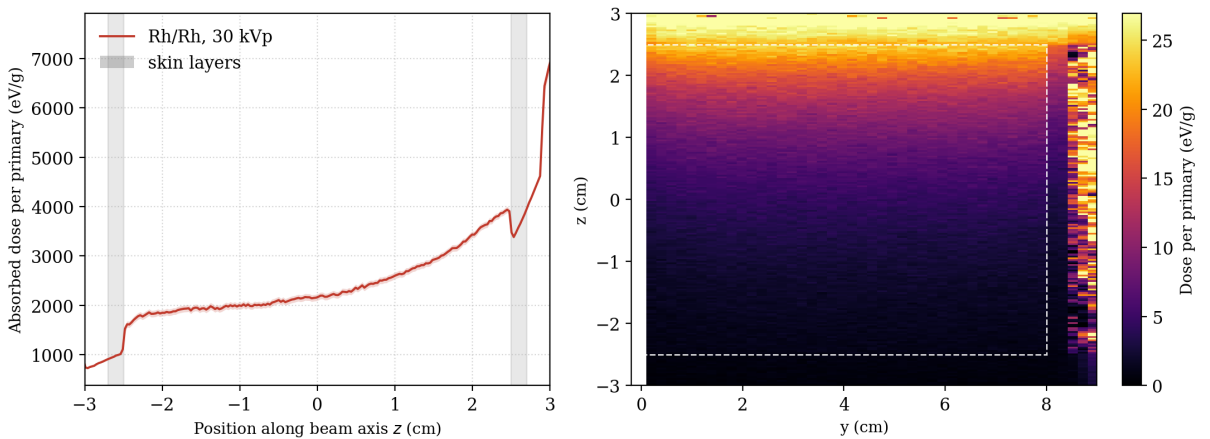
**Figure 6:** 2D dose maps in the  $yz$  plane through the centre of the breast (glandular 50%,  $x = 0$ ) for the three spectra. The dashed white box delimits the breast volume. The colour scale is identical in the three panels to allow direct visual comparison; in all cases the dose is highest near the entrance face ( $z = +3$  cm) and decreases with depth.

The expected qualitative trend is that softer spectra (Case 1) deposit a larger fraction of their energy close to the surface, since their lower mean energy means a shorter mean free path; the skin then absorbs a larger relative share. Harder spectra (Case 3) penetrate further into the breast and deposit more energy deep in the gland at the cost of a smaller skin contribution. The quantitative comparison of the three runs allows to verify and discuss this picture and to identify the spectrum that achieves the best “deep deposit / surface deposit” compromise for an average 50% breast.

## 4. Depth-Dose Distribution and Integrated Dose

The second part of the practice fixes the spectrum to one representative mammographic case (here we choose the Rh/Rh, 30 kVp beam, which corresponds to Run 3 in Table 2) and the glandularity to 50%, and asks for the depth-dose distribution inside the breast and the average dose received by the cylindrical breast volume.

### 4.1. Depth-dose profile and 2D dose map



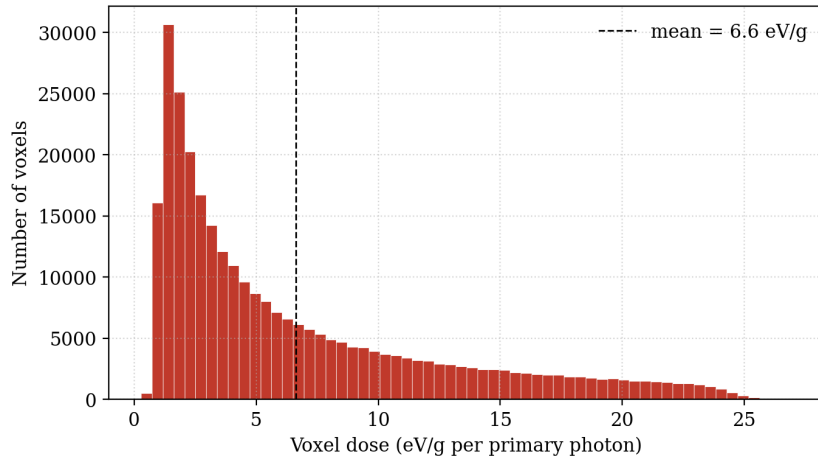
**Figure 7:** Dose distribution inside the breast for the Rh/Rh 30 kVp spectrum on glandular 50% tissue (half-cylinder height  $h = 5$  cm). *Left:* depth-dose curve  $D(z)$  along the beam axis; the gray bands mark the front and back skin layers ( $|z| \in [2.5, 2.7]$  cm). *Right:* 2D dose map in the  $yz$  plane through  $x = 0$ ; the dashed box delimits the breast volume.



The 1D depth-dose profile is obtained from `depth-dose.dat`, which contains the dose averaged over each  $z$ -slice of the dose grid. The left panel of Figure 7 shows  $D(z)$  from the entrance face of the breast ( $z \approx +2.5$  cm) down to the support plate. To support the 1D profile, the right panel of the same figure displays the 2D dose map in the  $yz$  plane through the breast axis ( $x = 0$ ), giving a complete view of the spatial distribution of the deposited energy.

The dose decreases approximately exponentially from the beam-entrance side towards the support. A shallow build-up region of a few millimetres can be present immediately below the skin, where the secondary electron equilibrium is being established; the 2D map highlights how the conical beam geometry produces a slight transverse spreading of the dose.

To complete the picture, the dose-volume distribution (Figure 8) is built from `3d-dose-map.dat` by binning all the breast voxels by their absorbed dose. It quantifies the dose heterogeneity inside the breast volume, which is hidden by the simple 1D profile.



**Figure 8:** Voxel-by-voxel distribution of the absorbed dose inside the breast volume (glandular 50%, Rh/Rh 30 kVp). The vertical line marks the mean,  $\langle D \rangle \approx 6.6$  eV/g per primary photon. The right-tail at high dose comes from the voxels close to the entrance face of the cylinder; the left-tail at low dose comes from the back of the breast.

## 4.2. Total dose to the breast

To convert the simulation output into a physical dose value, the energy deposited per primary photon must be scaled by the number of primary photons emitted by the tube during the exposure. The tube is assumed to operate at  $i = 100$  mA for  $t = 1$  s. The photon yield per mAs at the detector position depends on the chosen spectrum and is read from the SpekPy output. Combining these factors, the average dose to the breast volume is

$$\langle D \rangle = \frac{N_\gamma \langle \varepsilon \rangle}{m_{\text{breast}}}, \quad (2)$$

where  $\langle \varepsilon \rangle$  is the energy deposited per primary photon obtained from PENELOPE and  $m_{\text{breast}}$  is the mass of the cylindrical breast volume.

In the geometry provided with the practice the breast is modelled as a *half-cylinder*, with the patient's body filling the half-space behind it. The irradiated breast volume is therefore  $V = \frac{1}{2} \pi r^2 h$ , and the corresponding mass for glandular 50% tissue ( $\rho = 0.9819$  g/cm<sup>3</sup>) is

$$m_{\text{breast}} = \frac{1}{2} \pi r^2 h \rho = \frac{1}{2} \cdot \pi \cdot 64 \cdot 5 \cdot 0.9819 \approx 493.5 \text{ g}. \quad (3)$$

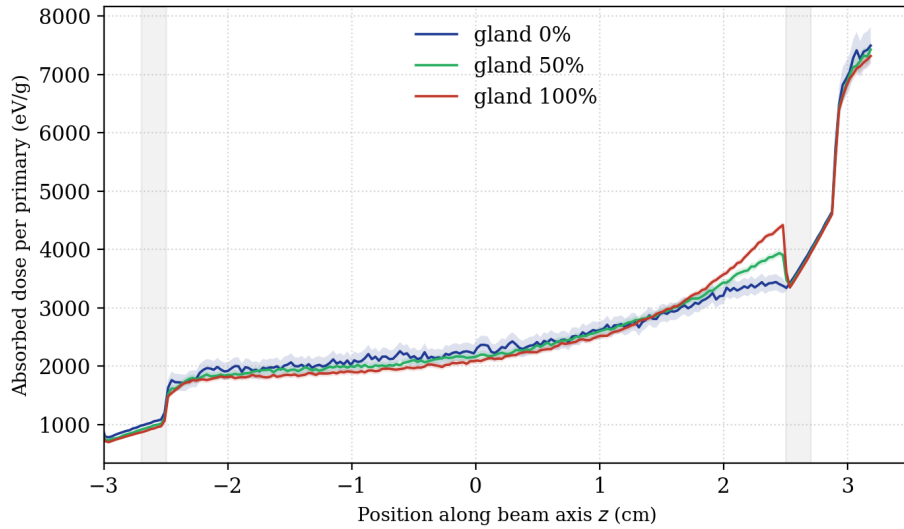
Substituting the simulation result for  $\langle \varepsilon \rangle$  and the photon flux gives the average dose reported in Table 4.

| Quantity                                                    | Value                                           |
|-------------------------------------------------------------|-------------------------------------------------|
| SpekPy fluence at $d = 100$ cm, $\Phi_\gamma$ (Rh/Rh)       | $2.27 \times 10^7$ photons/cm <sup>2</sup> /mAs |
| Beam-cone area at $d = 100$ cm                              | 977 cm <sup>2</sup>                             |
| Total photons in 100 mAs in the cone, $N_\gamma$            | $2.22 \times 10^{12}$                           |
| $\langle \varepsilon \rangle$ per primary photon (PENELOPE) | $3363.8 \pm 1.1$ eV                             |
| Breast mass, $m_{\text{breast}}$ (half-cyl, gland 50%)      | 493.5 g                                         |
| Total energy deposited                                      | $1.20 \times 10^{-3}$ J                         |
| <b>Average dose, <math>\langle D \rangle</math></b>         | <b>2.43 mGy</b>                                 |

Table 4: Total dose to the breast for the Rh/Rh, 30 kVp spectrum at 100 mAs, with 50% glandular tissue ( $h = 5$  cm half-cylinder).

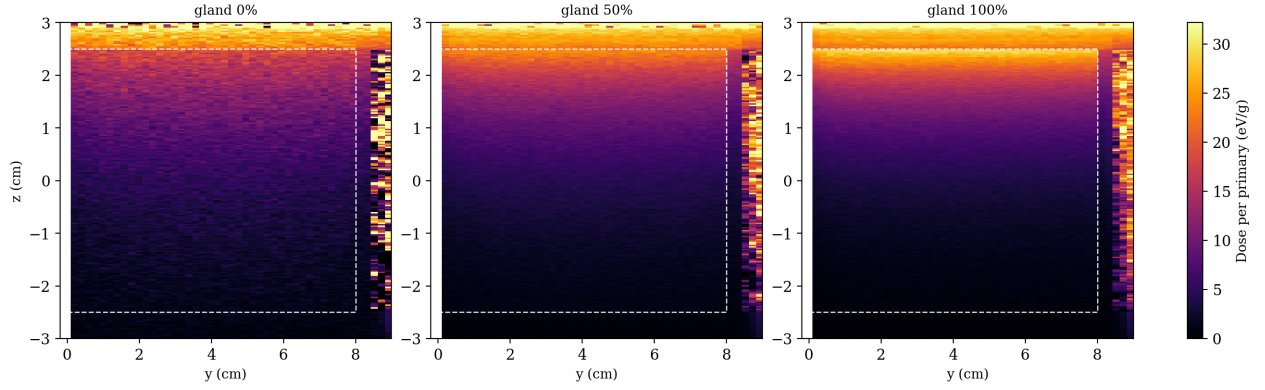
## 5. Effect of the Glandular Fraction

The last part of the practice repeats the depth-dose study of Section 4 for the two extreme tissue compositions, fully adipose (0% glandularity,  $\rho = 0.93$  g/cm<sup>3</sup>) and fully glandular (100%,  $\rho = 1.04$  g/cm<sup>3</sup>), and compares them with the 50% baseline. The X-ray spectrum is the same Rh/Rh 30 kVp used in Section 4. The two principal observables — 1D depth-dose profile and 2D dose maps — are presented together in Figure 9 and 10.



**Figure 9:** Comparison of the depth-dose profiles for the three glandular fractions at the same Rh/Rh, 30 kVp spectrum (all runs use the assignment-specified  $h = 5$  cm geometry). Denser tissues attenuate more rapidly with depth and absorb a higher dose near the entrance face.

The behaviour visible in Figure 9 matches the expectation from the mass attenuation coefficients of Figure 2: the three tissues differ modestly in  $\mu/\rho$  and by about 12% in density between the 0% and 100% extremes, but the two effects act in the same direction and combine into a sizeable  $\sim 27\%$  excess of the glandular 100% curve over the adipose one at the entrance face. The density ratio alone would account for about 12%; the rest comes from the higher effective atomic number of glandular tissue (more carbon and oxygen, less hydrogen), which strengthens the photoelectric contribution in the mammographic energy range. Figure 10 confirms this picture spatially: the bright top stripe of the breast volume sharpens and gains intensity from the adipose to the fully-glandular case, while the darker interior remains essentially unchanged.

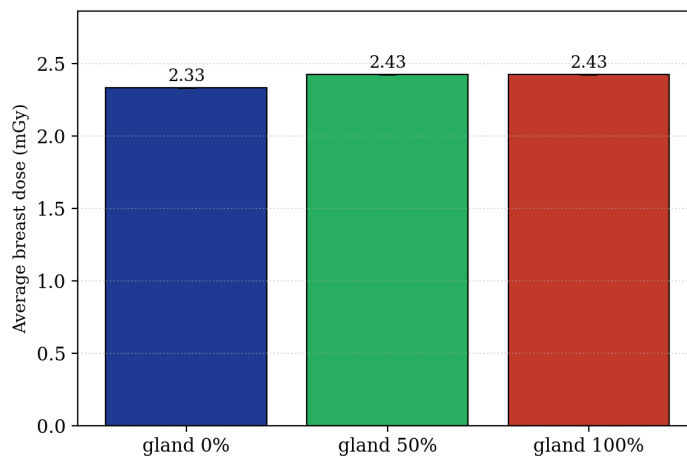


**Figure 10:** 2D dose maps in the  $yz$  plane through the centre of the breast for the three tissue compositions, all with the Rh/Rh, 30 kVp beam. The dashed box delimits the breast volume; the colour scale is identical in the three panels.

The integrated dose received by the breast at 100 mAs is reported in Table 5 and visualised in Figure 11. The dose grows monotonically with the glandular fraction; remarkably, the 50% and 100% values are within statistical uncertainty of each other, because the increase of  $\langle \varepsilon \rangle$  with the glandular fraction is largely compensated by the corresponding increase of the irradiated mass.

| Composition    | $\rho$ (g/cm <sup>3</sup> ) | Mass (g) | $\langle \varepsilon_{\text{br}} \rangle$ (eV/ $\gamma$ ) | $\langle D \rangle$ (mGy) |
|----------------|-----------------------------|----------|-----------------------------------------------------------|---------------------------|
| Glandular 0%   | 0.93010                     | 467.5    | $3066.7 \pm 2.3$                                          | $2.33 \pm 0.002$          |
| Glandular 50%  | 0.98190                     | 493.5    | $3363.8 \pm 1.1$                                          | $2.43 \pm 0.001$          |
| Glandular 100% | 1.04000                     | 522.8    | $3565.9 \pm 0.9$                                          | $2.43 \pm 0.001$          |

Table 5: Average dose received by the irradiated breast volume (half-cylinder,  $h = 5$  cm,  $r = 8$  cm) for the three tissue compositions at 100 mAs with the Rh/Rh, 30 kVp spectrum.



**Figure 11:** Average breast dose at 100 mAs as a function of the glandular fraction (Rh/Rh, 30 kVp). The dose grows monotonically with the glandular content; the 50% and 100% values are within statistics because the increase of  $\langle \varepsilon \rangle$  with the glandular fraction is largely compensated by the corresponding increase of the irradiated mass.

## Master summary

Table 6 collects the key quantitative results of the five PENELOPE runs in a single view, allowing a final cross-check of how the average dose, the breast/skin partition and the depth profile react to the two physical levers studied (spectrum hardness and glandular fraction).

| Run | Spectrum      | Gland. % | $\langle \varepsilon_{\text{br}} \rangle$ (eV/ $\gamma$ ) | $\langle \varepsilon_{\text{sk}} \rangle$ (eV/ $\gamma$ ) | $\langle D \rangle$ (mGy) |
|-----|---------------|----------|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------|
| 1   | Mo/Mo, 25 kVp | 50       | 2926.6                                                    | 817.2                                                     | 1.14*                     |
| 2   | Mo/Rh, 29 kVp | 50       | 3418.1                                                    | 816.9                                                     | 2.05*                     |
| 3   | Rh/Rh, 30 kVp | 50       | 3555.0                                                    | 803.0                                                     | 2.43                      |
| 4   | Rh/Rh, 30 kVp | 0        | 3066.7                                                    | 387.4                                                     | 2.33                      |
| 5   | Rh/Rh, 30 kVp | 100      | 3565.9                                                    | 304.2                                                     | 2.43                      |

Table 6: Master summary of the five Monte Carlo runs.  $\langle \varepsilon_{\text{br}} \rangle$  and  $\langle \varepsilon_{\text{sk}} \rangle$  are the energy deposited per primary photon in body 1 (breast) and body 2 (skin), respectively.  $\langle D \rangle$  is the average breast dose for an exposure of 100 mAs. \*Runs 1 and 2 use the  $h = 6$  cm geometry ( $m_{\text{breast}} = 592$  g for gland 50%); the harder is the beam, the higher the dose at fixed exposure, both because of the increased fluence and the increased mean photon energy.

## 6. Conclusions

The Monte Carlo simulations performed in this practice illustrate the two main physical levers that govern the dose deposited in the breast during a mammographic exposure: the *energy spectrum* of the incoming X-ray beam and the *tissue composition* of the breast itself. Softer spectra such as Mo/Mo 25 kVp deposit relatively more energy close to the surface, raising the skin contribution and producing the most marked depth-dose attenuation; harder beams such as Rh/Rh 30 kVp penetrate deeper and reach the rear of the breast volume more uniformly, but at the price of a lower local photoelectric contrast. The Mo/Rh, 29 kVp spectrum sits in between and is consistently used for breast thicknesses in the 5–7 cm range, the case considered in this study.

The variation of the breast composition in the 0%–100% glandularity range affects the depth-dose curve through two correlated mechanisms, the change of mass density and the change of effective atomic number, both of which act in the same direction. The fully-glandular breast ( $\rho = 1.04$  g/cm<sup>3</sup>) absorbs the largest fraction of the beam and concentrates the dose closer to the entrance face, while the fully-adipose breast ( $\rho = 0.93$  g/cm<sup>3</sup>) offers a lower attenuation and a flatter depth profile. The corresponding average dose at 100 mAs grows from 2.33 mGy at 0% to 2.43 mGy at 100% glandularity, remaining within the few-mGy range typical of clinical mammography. Notably, the 50% and 100% doses are within statistical uncertainty of each other: the increase in  $\langle \varepsilon \rangle$  is nearly compensated by the increase of the irradiated mass. The cross-comparison summarised in Table 6 allows to read at a glance how each of the two physical levers maps into the average breast dose and the breast-to-skin partition.